



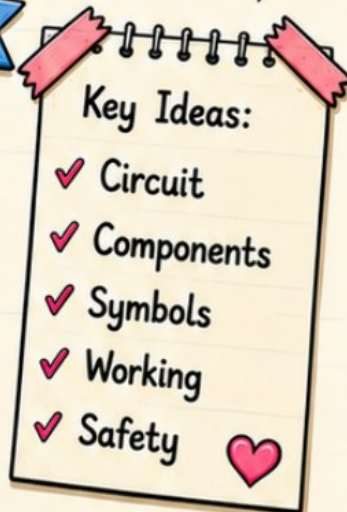
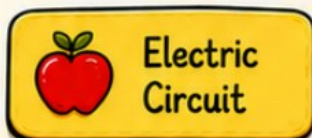
CLASS 7

CHAPTER 3



ELECTRICITY- CIRCUIT AND THEIR COMPONENTS

UNDERSTAND • CONNECT • ILLUMINATE



Science Creative Notes





Chapter 3: Electricity - Circuits and their Components



1. Uses of Electricity

Electricity is an **important** part of our daily life. It is used for:

- **Cooking** – kettle, mixer and grinder
- **Lighting** – homes, offices and factories
- **Transportation** – trains, trams and buses
- **Heating and Cooling** – fans, heaters, refrigerators and air conditioners
- **Entertainment** – television and radio
- **Communication** – mobile phones, computers and Internet
- **Other uses** – water pumps and cranes



2. Electricity from Water



- **Hydroelectric power** is generated using the energy of **flowing or falling water**.
- Dams such as **Bhakra Nangal Dam** are used to generate electricity.



Activity - Let Us List



Look around your house and list any five electrical appliances and their uses.



Appliance		Use
Fan		Cooling
Television		Entertainment
Refrigerator		Cooling and storing food
Computer		Work and communication
Mixer		Grinding and mixing



3. Electrical Safety



- Electricity is useful but can be **dangerous** if handled carelessly.
- Never touch electrical appliances with **wet hands**.
- Do not touch **bare or damaged** wires.
- Keep electrical appliances **away from water**.
- Do not insert objects into **electrical sockets**.
- Electrical repairs should be done only by a **qualified person**.



3.1 A Torchlight



- A torchlight (flashlight) is a device that produces light using **electricity**.
- It has a **switch** that controls whether the lamp glows or not.



Activity 3.1 – Let Us Explore

- Observe the torchlight carefully. Notice its **lamp** and **switch**.
- Slide the **switch** and observe what happens to the lamp.
- In one position, the **lamp glows**, while in the other position, it does not.
- Open the torchlight and observe its inside. You will find **two or more electric cells**.



Why does the torch lamp glow when the switch is in one position?

Observation:



When the switch is **ON**, the torch glows because the circuit is complete and electric current can flow through it. When the switch is **OFF**, the circuit is broken, so the torch does not glow.



Conclusion: A complete circuit is needed for electric current to flow.

3.2 A Simple Electrical Circuit



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3.2.1 Electric Cell

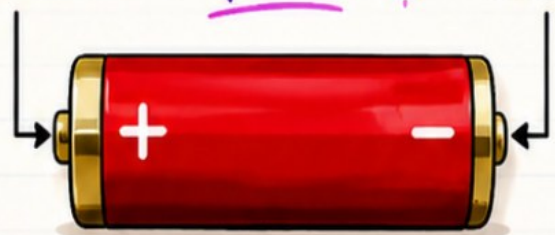


- An electric cell is a source of electrical energy.
- A cell has two terminals:
 - **Positive terminal (+)**
 - **Negative terminal (-)**
- The positive terminal is usually marked with a **+** sign, while the negative terminal is marked with a **-** sign.

Positive terminal

EduPath IQ

Negative terminal



Activity 3.2 – Let Us Observe

- Take an electric cell and observe its two terminals.
- Notice the positive and negative terminals carefully.



Observation:

An electric cell has two terminals: a positive terminal (metal cap) and a negative terminal (metal disc). Those terminals help in the flow of electric current. The electric cell acts as a portable source of electrical energy.



3.2.2 Battery



- A battery is made by connecting two or more electric cells.
- The **positive terminal** of one cell is connected to the **negative terminal** of the next cell. Such a combination of two or more cells is called a **battery**.
- The cells in a battery can be connected in different ways.
- When cells are connected in the correct order, they can provide **more electrical energy**.

Battery (2 cells)



or



or



Activity 3.3 – Let Us Experiment



- Take a torch that uses two cells.
- Open its cell compartment and observe how the cells are placed.
- Try reversing the position of the cells and check whether the torch glows.
- Place the cells back in the correct position and observe the torch again.



Observation and Conclusion:

Observation: The torch glows when the cells are placed in the correct order.
When the cells are placed incorrectly, the torch does not glow.



Conclusion: Cells need to be connected in the correct order for the torch to work.



Battery Pack



- Many devices need more than one cell.
- Two or more cells **connected together** form a battery.
- Connecting more cells can provide **more electrical energy** and/or make the current flow for a longer time.



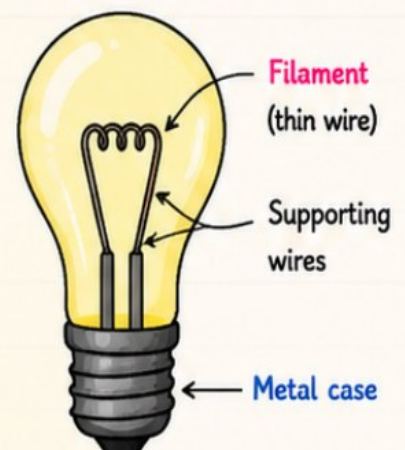
3.2.3 Electric Lamp



Incandescent Lamp

- An incandescent lamp has a thin wire called a **filament** inside a glass bulb.
- The filament is connected to two supporting wires.
- The supporting wires are connected to the **metal case** at the bottom.
- When electric current passes through the filament, it becomes **very hot** and **glows**, producing light.
- The **glass bulb** protects the filament from direct contact with air.

Incandescent Lamp



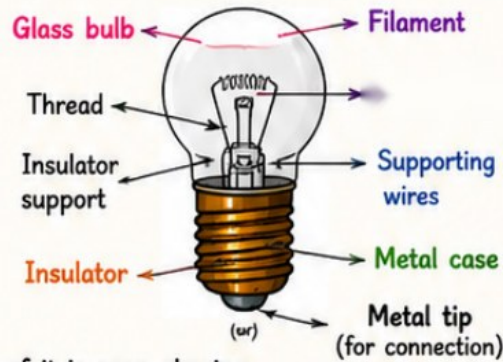


Activity 3.4 – Let Us Observe



Observe an incandescent lamp carefully and identify:

- Glass bulb
- Filament
- Supporting wires
- Metal case
- Insulator



Switch on the torch and observe which part of it is seen glowing.

Observation:



The **filament glows** when electric current passes through it.



Conclusion:

The filament of an incandescent lamp produces light when it becomes **hot** due to the flow of electric current.



LED Lamp

- Many modern torches use an **LED (Light Emitting Diode)** lamp instead of an incandescent lamp.
- LED lamps are commonly used in **torches** and many other **devices**.



Activity 3.5 – Let Us Observe

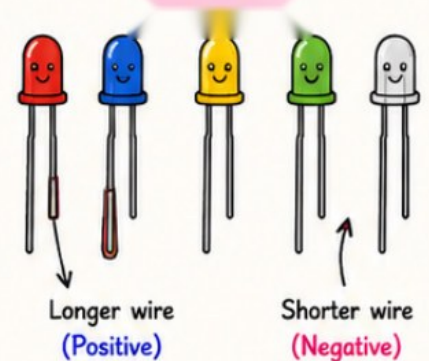


- Take an LED of any colour and carefully observe its shape.
- Notice the two wires (**terminals**) attached to the LED.
- One wire is longer than the other.

Observation:



The LED has two terminals. The longer wire is the **positive terminal**, while the shorter wire is the **negative terminal**.



Conclusion:

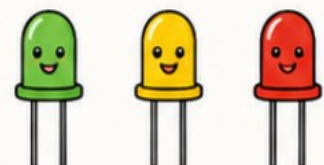
An LED has two terminals, and it **glows** when connected to a circuit in the **correct direction**.



LED vs Incandescent Lamp



- Unlike an incandescent lamp, an **LED** does not have a filament.
- It has two terminals, one **positive** and one **negative**.
- LEDs are available in different **shapes** and **colours**.



3.2.4 Making an Electric Lamp Glow Using an Electric Cell or Battery

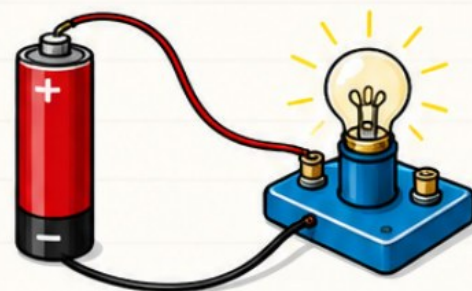


Activity 3.6 – Let Us Construct

Aim: To make an electric lamp glow using an electric cell.

Procedure

- Take an **electric cell** and an **incandescent lamp** used in a torch.
- Remove about **1 cm of plastic covering** from both ends of two wires.
- Attach the wires to the **ends of the cell holder**.
- Connect the other ends of the wires to the **lamp holder**.
- Fix the lamp in the holder and complete the connections.

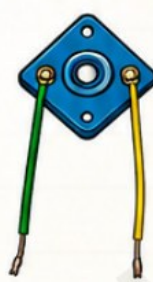


Let Us Explore

Make different arrangements of the cell, wires and lamp as shown in the figures.

Before connecting each arrangement:

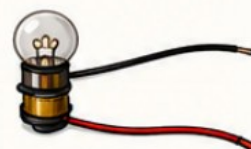
- Predict whether the lamp will glow or not.
- Connect the circuit and observe.
- Compare your prediction with the observation.



(a)



(b)



(c)



Table 3.1 – Trying to Make the Lamp Glow



S.No.	Arrangement of Cell and Lamp	Prediction	Observation
1.		The lamp will glow.	The current will flow as all connections are properly connected and the circuit is complete.
2.		The lamp will not glow.	The circuit is incomplete, so the current will not flow.
3.		The lamp will not glow.	The circuit is incomplete due to the disconnected wire.
4.		The lamp will not glow.	Two wires of the bulb are connected to the positive terminal of the cell.
5.		The lamp will not glow.	Two wires of the bulb are connected to the negative terminal of the cell.
6.		The lamp will glow.	The current will flow as all connections are properly connected and the circuit is complete.





Observation and Conclusion:

- The lamp glows only when the **electric circuit is complete**.
- If the circuit is incomplete, the lamp **does not glow**.
- The colour of the lamp's glow (**brightness**) may also be different in different lamps.



Conclusion: An electric lamp glows when there is a **complete path** for electric current to flow.

3.2.5 An Electrical Circuit



- An electrical circuit is a **complete path** through which electric current can flow.
- When the two terminals of an electric cell are **connected properly** to the two terminals of a lamp, the **lamp glows**.
- If the circuit is incomplete, the lamp **does not glow**.
- In an incandescent lamp, current passes through the **filament**, making it **hot** and causing it to **glow**.



Fascinating fact



Sometimes an incandescent lamp may glow even when connected to a cell with a very low current, because its filament has a **high resistance**.



Activity 3.7 – Let Us Experiment

Aim

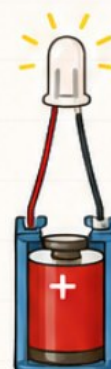
To make an LED glow using electric cells.

Procedure

- Take **two electric cells**, an **LED**, a **cell holder** and **two wires**.
- Connect the wires to the cell holder and insert the cells in the correct direction.
- Connect the other ends of the wires to the **LED**.
- Try **connecting** the LED in both directions and observe whether it glows.



(a)

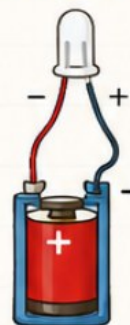


(b)

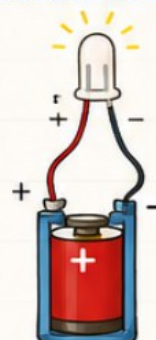


Observation:

- The LED glows in the first arrangement (Fig. 3.10c) but does not glow in the other arrangement (Fig. 3.10d).
- This is **because** electric current can pass through an LED in **one direction only**.
- The LED glows only when its **positive terminal** (longer wire) is connected to the **positive terminal** of the battery, and the **negative terminal** (shorter wire) is connected to the negative terminal of the battery.
- When current passes through the LED, it glows.



(a) Does not glow



(b) Glows



Observation and Conclusion

- The lamp glows only when the **electric circuit is complete**.
- If the circuit is incomplete, the lamp **does not glow**.
- The **brightness** of the lamp may also be different in different lamps.

Conclusion:

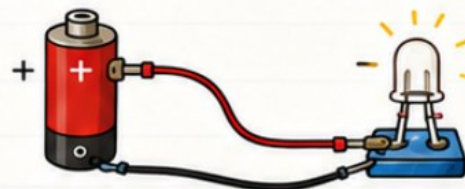
An electric lamp glows when there is a **complete path** for electric current to flow.



Important Point

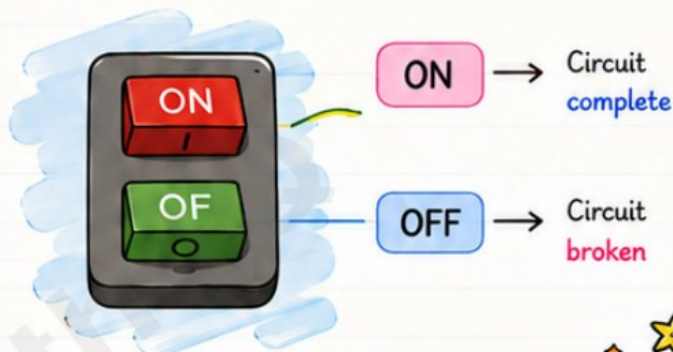
- Longer wire of LED → **Positive (+)**
- Shorter wire of LED → **Negative (-)**

The positive and negative terminals of the battery must be connected to the corresponding terminals of the LED.



3.2.6 Electric Switch

- An electric switch is a device used to complete or break an electric circuit.
- A switch can be placed anywhere in a circuit.
- When the switch is **ON**, the circuit is **complete** and electric current flows.
- When the switch is **OFF**, the circuit is **broken** and electric current does not flow.



Activity 3.8 – Let Us Construct

Aim: To make a simple electric switch.

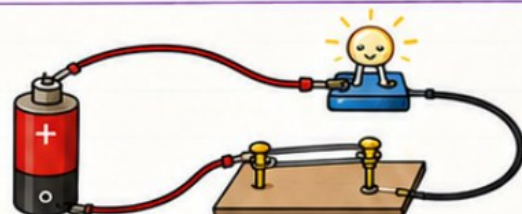
- Take two **drawing pins**, a **safety pin**, a **paper clip** and a small piece of cardboard.
- Fix the drawing pins on the cardboard.
- Insert the safety pin through one drawing pin so that it can rotate freely.
- Fix the second drawing pin so that the free end of the safety pin can touch it.
- Connect a **wire** to each drawing pin.

Conclusion: A simple switch can be made using a safety pin and drawing pins. It can be used to make or break an electric circuit.

- Fix the drawing pins on the cardboard.
- Insert safety pin through one drawing pin.
- Let it rotate freely.
- Fix the second drawing pin.
- Connect a wire to each drawing pin.

Activity 3.9 – Let Us Test

- Connect the electric cell, lamp and switch in a circuit.
- Rotate the safety pin so that it touches both drawing pins.



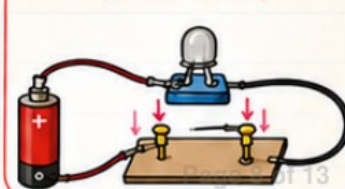
Observation and Conclusion

- When the safety pin touches both drawing pins, the circuit is **complete** and the lamp **glows**.
- When the safety pin does not touch both drawing pins, the circuit is **broken** and the lamp **does not glow**.

(Circuit complete)



(Circuit broken)

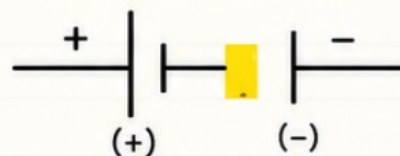




Important Point

- In the symbol for an electric cell, the **long line** represents the **positive (+) terminal**.
- The **short line** represents the **negative (-) terminal**.

Electric Cell Symbol



LED Symbol

- In the symbol for an LED, the **triangle** points in the direction in which the current flows.
- The **two arrows** pointing outwards show that the LED emits light.

LED Symbol

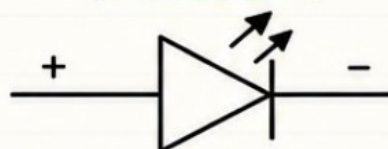
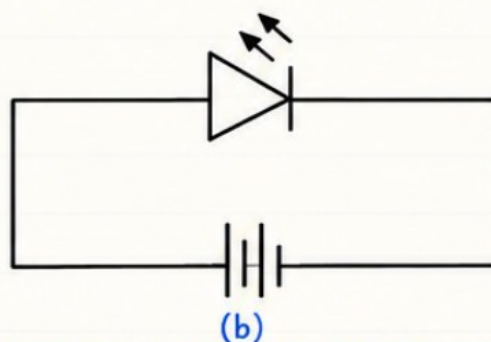
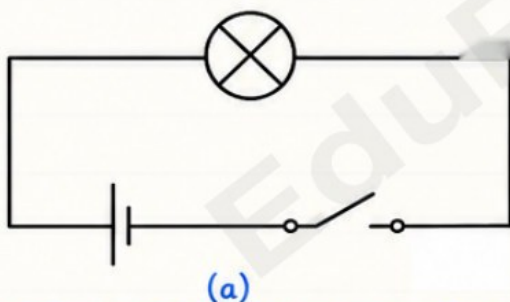


Fig. 3.12 Positive and negative terminals in the symbols of cell and LED.



Activity 3.10 – Let Us Draw

- Use the standard symbols to draw the circuit diagrams shown in the textbook.
- Draw the circuit using an **electric cell**, **switch** and **electric lamp**.
- Also draw a circuit using a **battery** and **LED**.



Observation and conclusion: The circuit diagram can be drawn easily with the help of symbols of different electric components.



Dive Deeper – Standard Symbols

- International organizations such as the **International Electrotechnical Commission (IEC)**, **American National Standards Institute (ANSI)** and **Institute of Electrical and Electronics Engineers (IEEE)** create standard symbols for electrical and electronic parts.
- Using the same symbols across the world helps people from different countries and industries **understand electrical diagrams easily**.



3.4 Electrical Conductors and Insulators



- ★ Materials used for making electric circuits may be **conductors** or **insulators**.
- ★ **Conductors** allow electric current to pass through them.
- ★ **Insulators** do not allow electric current to pass through them easily.

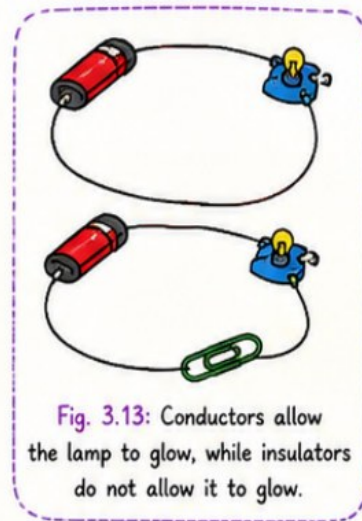


Fig. 3.13: Conductors allow the lamp to glow, while insulators do not allow it to glow.

Activity 3.11 – Let Us Investigate

Aim To find out which materials allow **electric current** to pass through them.

Procedure



- 1 Connect an **electric cell** and a **lamp** in a circuit.
- 2 Leave a small gap between the two ends of the wire.
- 3 Touch the two ends of the wire with an object and observe whether the lamp glows.
- 4 Test different objects such as **metal**, **plastic**, **wood**, **glass**, **paper**, **rubber** and **pencil lead**.



Electric current flows only through certain materials!



Observation and Conclusion

- If the **lamp glows**, the material **allows electric current** to pass through it.
- If the lamp **does not glow**, the material does not allow electric current to pass through it easily.



Identifying Conductors and Insulators

Object	Material	Lamp Glows?	Conclusion
Stick	Wood	No	Insulator
Scale	Plastic	No	Insulator
Bangle	Glass	No	Insulator
Paper strip	Paper	No	Insulator
Candle	Wax	No	Insulator
Key	Metal	Yes	Conductor
Eraser	Rubber	No	Insulator



Remember: Metals are generally **conductors**, while non-metals are **insulators**.





Conductors

- Materials that **allow electric current to pass** through them are called electrical conductors.
- Metals are generally **good** conductors of electricity.
- Examples:** copper, aluminium, iron and steel.



Copper



Aluminium



Iron / Steel



Insulators

- Materials that **do not allow** electric current to pass through them **easily** are called electrical **insulators**.
- Examples:** wood, plastic, glass, paper, rubber and wax.



Wood



Plastic



Glass



Paper



Rubber



Wax



Best Electrical Conductors

- Silver, copper** and **gold** are the best electrical conductors.
- Copper** is mainly used for making electrical wires because it is comparatively low-cost and available in abundance.
- Different types of electrical wires are used for different purposes.



Types of Electrical Wires



- Single-core wire (used in homes)



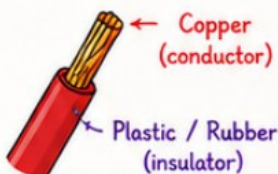
- Multi-strand wire (used in appliances)



- Armoured wire (used in industries)



Why Are Wires Covered?



Copper (conductor)

Plastic / Rubber (insulator)

- Plastic, rubber and ceramics are electrical insulators.
- Electrical wires, switches, connectors, plugs and sockets contain conductors.
- Insulators such as rubber, plastic and ceramics are used to cover wires, plugs and switches to protect people from electric shocks.



Electrical Safety



- Our body is a conductor of electricity.



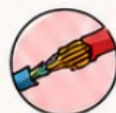
- Electric current passing through our body can cause severe injury or even death.



- Never touch switches or plugs with wet hands.



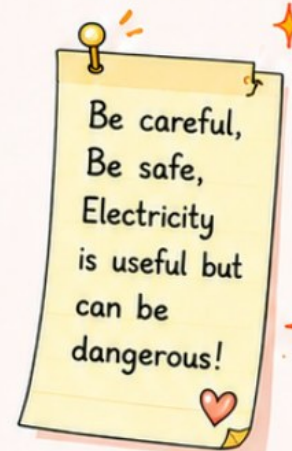
- Do not use electrical devices in wet areas.



- Never handle equipment with damaged insulation or broken plugs.



- Do not overload sockets or use too many devices at the same time.



Remember: **Conductors** allow current to pass, **insulators** do not. Use electricity wisely and always follow **safety rules**!





Dive Deeper – DC and AC



- Electricity from cells and batteries is called **Direct Current (DC)**.
- DC is generally used for **small devices**.
- Electricity supplied through wall sockets comes from power plants and is called **Alternating Current (AC)**.
- AC can be used to run **larger appliances**.

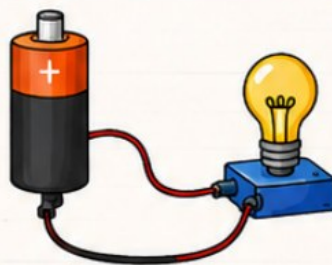


Remember

- Cell/Battery → **DC** → Small devices
- Wall socket → **AC** → Larger appliances



DC (Direct Current)



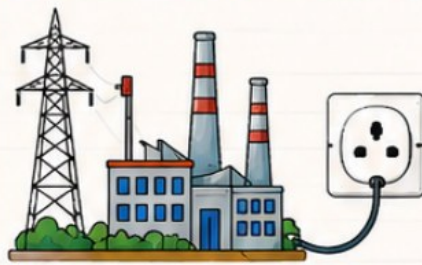
Comes from cells and batteries.

Used in:

small devices



AC (Alternating Current)



Comes from power plants through wall sockets.

Used in:

larger appliances



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